ProGrade: A Medallion Architecture-Driven Early Warning Framework for Academic Risk Prediction Using Ensemble Learning and Temporal Checkpoints

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***Abstract*—Early identification of at-risk students remains a persistent challenge in higher education, where interventions are frequently applied after the window for effective remediation has closed. This paper presents ProGrade, an end-to-end early warning framework integrating a Medallion data warehouse architecture with temporal checkpoint modeling and explainable machine learning. Three public educational datasets are unified within a Kimball star-schema pipeline (Bronze, Silver, and Gold layers). A temporal feature-masking protocol progressively reveals features at simulated Weeks 4, 8, and 12, enabling systematic evaluation of the accuracy–lead time tradeoff. To ensure methodological validity, Mutual Information feature selection, SMOTE oversampling (k-neighbors$=5$), and feature scaling are applied exclusively within training folds, preventing data leakage. We benchmark Logistic Regression, Random Forest, and XGBoost across all checkpoints. At Week 12, XGBoost models achieve an F1-Macro of 0.852 ± 0.005 and an ROC-AUC of 0.920 on held-out test data (confirmed via five-seed stability analysis). Furthermore, models at Week 8 achieve an F1-Macro over 0.80, offering significant lead time for counselor interventions. SHAP-based feature attribution delivers both global and individualized explanations, providing three-tier transparency that facilitates actionable, per-student intervention recommendations.**

***Index Terms*—Educational Data Mining, Student Dropout Prediction, Data Warehouse, Medallion Architecture, SHAP Explainability, Ensemble Learning, Early Warning Systems, Data Leakage Prevention.**

I. Introduction

Student attrition in higher education carries profound economic and social implications for institutions, labor markets, and individuals [1]. Worldwide, dropout rates continue to burden institutional resources, denying students the educational outcomes vital to their long-term social mobility [2]. Despite the high costs associated with student attrition, the standard institutional response remains functionally reactive: students take on an "at-risk" label only after final grades are recorded, which is critically late to offer meaningful interventions along their academic pathways.

Machine learning (ML) has emerged as a highly promising solution within Educational Data Mining (EDM) to predict student outcomes [3], [4]. Ensemble approaches—such as Random Forest (RF) [5] and gradient-boosted trees like XGBoost [6]—consistently display strong predictive performance on structured educational data [7], [8]. Nevertheless, the transition from research prototypes to deployable institutional tools is heavily impeded by four common shortcomings in the extant literature:

1) **Lack of reproducible data architectures.** Most EDM studies apply ML algorithms to isolated datasets or flat files lacking formalized data governance [3]. Data pipelines built on established dimensional modeling principles [9] are rarely utilized, despite being the industry standard for managing analytical workloads [10].
2) **Data leakage in preprocessing pipelines.** Many research efforts perform SMOTE oversampling, normalization, or imputation *prior* to the train-test split, resulting in exaggerated performance measures that fail to generalize to unobserved data [11], [12].
3) **Temporal drift between prediction and intervention.** Predictive systems typically assess risk either at enrollment (relying on limited pre-admission features) or at the end of the semester (using outcome variables). The crucial mid-semester timeframe—a historically under-studied period wherein behavioral signals accumulate before final academic outcomes manifest—is rarely capitalized upon [13], [14].
4) **Insufficient model interpretability.** Academic advisors must provide clear, auditable rationales behind targeted interventions. Opaque "black-box" predictions fundamentally fail to cultivate the institutional trust necessary for widespread adoption [15], [16].

To bridge these gaps, we introduce **ProGrade**—a production-oriented analytics framework combining a Medallion data warehouse with leakage-free ensemble modeling, temporal checkpoint assessment, and SHAP-based explainability.

A. Research Questions

- **RQ1:** What is the impact of a structured Medallion warehouse architecture (Bronze $\rightarrow$ Silver $\rightarrow$ Gold) and temporal feature masking on the reproducibility and predictive quality of EDM models?
- **RQ2:** What is the most operationally viable early-warning tradeoff between prediction accuracy and intervention lead time?
- **RQ3:** To what extent can SHAP feature-attribution explanations support transparent, per-student risk stratification for academic counselors?

B. Contributions

C1. Enterprise Data Warehouse (EDW) with Medallion Architecture: We utilize a three-tiered Medallion Architecture (Bronze, Silver Star-Schema, Gold Data Mart) to perform educational analytics, establishing a fully reproducible and auditable data foundation.

C2. Temporal Checkpoint Protocol: We construct a feature-masking protocol that emulates progressive information availability at Weeks 4, 8, and 12. This allows for a principled quantification of the accuracy-intervention lead time tradeoff.

C3. Leakage-Free ML Pipeline: Our feature selection and preprocessing are strictly post-split. Specifically, Mutual Information feature selection, SMOTE (k-neighbors=5), and StandardScaler are applied exclusively to the training data, ultimately yielding an F1-Macro of 0.852 ± 0.005 on the held-out test data.

C4. Explainable Risk Stratification: We leverage SHAP TreeExplainer to produce global feature importance summaries alongside student-specific waterfall plots, which are annotated by a three-level risk scale (Low / Medium / High).

II. Related Work

A. Educational Data Mining and Prediction of Student Outcomes

Data mining has been widely applied within educational settings [3], [4]. Hellas et al. [8] conducted a systematic review of predictive modeling for academic performance, determining that feature engineering and model selection are the most critical determinants of accuracy. More recent literature [7], [17] confirms that, for tabular educational data, ensemble techniques—specifically Random Forest and gradient boosting—continue to outperform linear models and neural networks. In a recent review of dropout prediction methodologies, Umer et al. [18] highlight persistent gaps in temporal modeling and cross-institutional generalization.

B. Data Governance in Educational Analytics

Despite the sophistication and widespread adoption of dimensional modeling in industry [9], educational analytics pipelines largely remain ad-hoc. Abella-García et al. [10] analyzed data warehouse architectures in education, finding that a lack of standardized schemas significantly hinders institutional adoption. ProGrade bridges this gap by introducing a Medallion Architecture that separates raw data ingestion, dimensional modeling, and analytical outputs into distinct, explicit layers with clear data lineage.

C. Early Warning Systems

Early Warning Systems (EWS) for student risk have transitioned from rule-based heuristic models to machine learning-based predictions. Arnold and Pistilli [19] introduced Purdue's Course Signals, a pioneering framework that utilized simple traffic-light indicators. Subsequent applications expanded on this foundational work through open-source analytics [20] and early prediction at Week 4 based on standard grading [13]. However, standard systems typically evaluate risk at a single time point rather than systematically comparing prediction quality across a continuum of checkpoints. While Aulck et al. [14] evaluated student dropouts longitudinally across broader semesters, they did not investigate the temporal dynamics within a single semester. In contrast, Martins et al. [21] explored phased prediction on the UCI Dropout dataset, serving as a direct point of comparison for our checkpoint protocol.

D. Data Leakage and Methodological Validity

Class imbalance is inherent to dropout prediction, given that at-risk students consistently constitute a minority group [11]. The most popular mitigation technique is SMOTE [12]; however, its misapplication prior to data splitting introduces synthetic samples projected from test data, thereby severely polluting model evaluation [3]. ProGrade maintains a rigid structural separation: SMOTE oversampling and feature scaling are computed exclusively within the training partition, and validation/test sets are transformed strictly using parameters derived from this training data.

E. Explainable AI in Education

Khosravi et al. [15] detail how explainable artificial intelligence (XAI) is an absolute prerequisite for institutional adoption within educational spheres. Recently, Alwadi et al. [16] successfully demonstrated SHAP-based transparent student-supported early detection. The SHAP framework itself, developed by Lundberg and Lee [22] and grounded in cooperative game theory, provides mathematically sound global and local feature attribution. ProGrade utilizes the SHAP TreeExplainer to generate individualized waterfall plots, systematically identifying the specific driving features behind the risk classification assigned to each flagged student.

F. Complementary Modeling Approaches

Recent efforts have also examined alternative architectures for educational prediction. Graph Neural Networks (GNNs) and attention-based temporal models have been implemented to capture relational and sequential structures inherent in student interaction data [23]. Additionally, privacy-preserving cross-institutional analytics have been proposed via federated learning frameworks, which facilitate collaborative model

training without exposing raw student records [24]. These complementary approaches address relational dynamics and data privacy, whereas ProGrade specifically targets the optimization of structured tabular data governance, ensuring leak-free performance evaluation and interpretability.

III. Methodology

ProGrade comprises four integrated components: (1) a Medallion data warehouse pipeline, (2) a temporal feature-masking protocol, (3) a leakage-free model training and evaluation pipeline, and (4) a SHAP-based explainability module. Fig. 1 illustrates the Silver layer star-schema design.

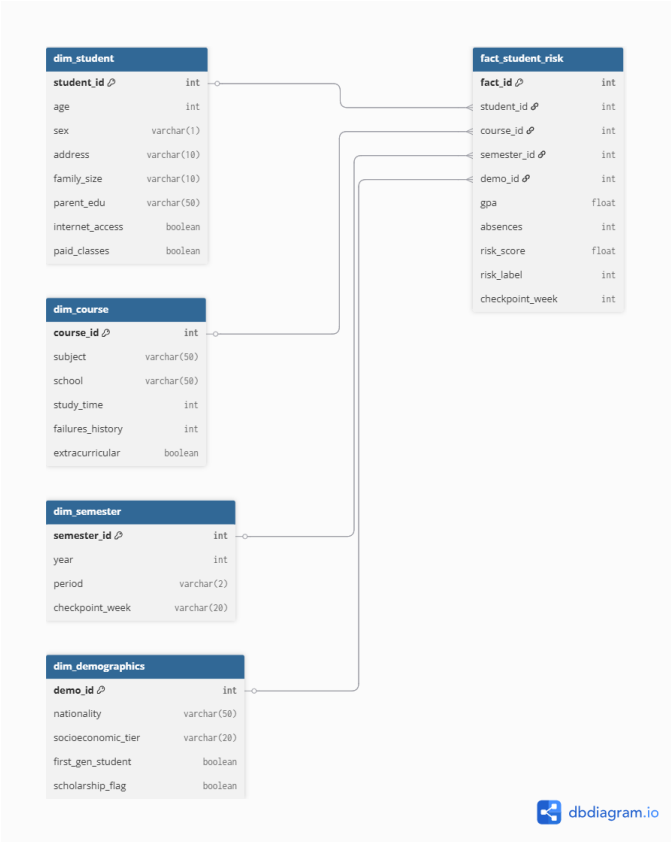


Fig. 1: ProGrade Silver layer star-schema. The central `FactStudentRisk` table links to four dimension tables (`DimStudent`, `DimCourse`, `DimSemester`, `DimDemographics`) via surrogate keys.

A. Datasets

Three publicly available educational datasets are unified within the ProGrade warehouse:

- **UCI Student Performance** [25]: 649 records of secondary school students in Portugal with demographic, social, and academic attributes. The binary risk label is derived as $G3 < 10$ (final grade below passing threshold).
- **UCI Dropout and Academic Success** [1], [26]: 4,424 records from a Portuguese polytechnic institution with 36 attributes spanning enrollment, demographic, academic, and macroeconomic dimensions. Risk label: `Target` = "Dropout". This is the primary modeling dataset due to its richness and temporal feature structure.
- **Kaggle Students Performance** [27]: 2,392 records with academic, behavioral, and parental attributes. Risk label: `GradeClass` ≥ 3 (grades D or F).

B. The Medallion Warehouse Pipeline

The ETL pipeline follows the Medallion Architecture pattern, decomposing data processing into three layers:

- **Bronze (Raw Ingestion):** Source datasets are ingested with path-integrity validation via declarative manifests. Raw files are stored without modification to preserve data provenance.
- **Silver (Dimensional Model):** Data is cleaned, imputed (median for numeric, mode for categorical), and mapped to a Kimball star-schema [9]. As shown in Fig. 1, the central `FactStudentRisk` table stores numeric measurements (GPA, absences, risk score) and links to four dimension tables via surrogate keys: `DimStudent` (personal/family attributes), `DimCourse` (subject/school context), `DimSemester` (temporal checkpoint), and `DimDemographics` (socioeconomic indicators). The warehouse is implemented in SQLite using SQLAlchemy ORM with indexed foreign keys.
- **Gold (Data Mart):** A denormalized analytical view (`student_risk_mart.csv`) joins all dimensions with human-readable mappings (e.g., "Urban"/"Rural", GPA brackets, attendance tiers) for direct consumption by BI dashboards and counselor reports.

C. Temporal Feature Masking

To simulate the progressive availability of information during a semester, we define three temporal checkpoints with distinct feature masks applied to the UCI Dropout dataset:

- **Week 4 (Enrollment):** Only demographic and pre-enrollment features are available: marital status, application attributes, parental qualifications and occupations, admission grades, age, gender, scholarship status, and nationality (21 features).
- **Week 8 (Mid-Semester):** First-semester academic performance features become visible: curricular units credited, enrolled, evaluated, approved, grades, and units without evaluations ($21 + 6 = 27$ features).
- **Week 12 (Late-Semester):** The full feature set is unlocked, including second-semester academic performance (GPA, curricular unit metrics) and macroeconomic indicators (unemployment rate, inflation, GDP) ($27 + 9 = 36$ features).

Columns used to derive the target variable (`Target`, `G3`, `GradeClass`) are explicitly excluded from the feature set at all checkpoints to prevent target leakage.

D. Leakage-Free Preprocessing and Model Training

Algorithm 1 summarizes the per-checkpoint training protocol. The pipeline enforces four leakage-prevention mechanisms:

Algorithm 1 Per-Checkpoint Training Protocol

Require: Dataset D, checkpoint week $w \in \{4, 8, 12\}$
1: $\mathbf{X} \leftarrow$ Apply feature mask $\mathcal{F}_w$ to D
2: One-hot encode categorical features in $\mathbf{X}$
3: Split: $\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{val}}, \mathbf{X}_{\text{test}}$ (70/15/15, stratified)
4: Compute MI scores $\text{MI}(\mathbf{X}_{\text{train}}, y_{\text{train}})$ on training data **only**
5: Select top-k features ($k = 15$) by MI ranking; subset all splits
6: Apply SMOTE ($k_{\text{neighbors}}$=5) to $(\mathbf{X}_{\text{train}}, y_{\text{train}})$ **only**
7: Fit StandardScaler on $\mathbf{X}_{\text{train}}$; transform all splits
8: Train models {LR, RF, XGBoost} on $\mathbf{X}_{\text{train}}$
9: Evaluate on $\mathbf{X}_{\text{test}}$ using F1-Macro, ROC-AUC, κ

1) **Target isolation:** Source columns for the risk label are removed before feature selection.
2) **Post-split feature selection:** Mutual Information [28] scores are computed exclusively on the training partition; validation and test sets are subsetted to the same selected features, preventing selection bias leakage.
3) **Post-split balancing:** SMOTE (original variant [12], $k_{\text{neighbors}}$=5) is applied exclusively to the training partition.
4) **Post-split scaling:** StandardScaler parameters are fit on the training partition only; validation and test sets are transformed using these learned parameters.

Feature selection retains the top-15 features per checkpoint based on MI ranking. Collinear features ($|r| > 0.85$) are removed prior to selection, eliminating 2–4 features per checkpoint depending on feature set size.

Models. Three classifiers are benchmarked:

- *Logistic Regression* (LR) [29]: baseline linear model (C=1.0, LBFGS solver, max 1000 iterations).
- *Random Forest* (RF) [5]: bagging ensemble (300 trees, max depth 15, balanced class weights).
- *XGBoost* [6]: gradient boosting (300 rounds, max depth 6, learning rate 0.1, subsample 0.8).

All hyperparameters are fixed *a priori* based on established literature recommendations; no hyperparameter tuning is performed on the validation set. The validation partition ($n = 15\%$) is reserved for monitoring generalization during development, while final metrics are reported exclusively on the held-out test partition ($n = 15\%$). To assess result stability, the full training and evaluation protocol is repeated across five random seeds (42, 123, 256, 512, 1024), with mean $\pm$ standard deviation reported.

E. Evaluation Metrics

We adopt F1-Macro as the primary metric because it equally weights precision and recall across both classes, penalizing poor minority-class performance [30]. Supplementary metrics include accuracy, macro-averaged precision and recall, ROC-AUC, and Cohen's κ [31].

F. Three-Tier Risk Stratification and Explainability

Predicted risk probabilities are mapped to a three-tier intervention framework:

$$\text{Tier}(\mathbf{x}) = \begin{cases} \text{Low Risk} & \text{if } P(\text{risk}|\mathbf{x}) < 0.30 \\ \text{Medium Risk} & \text{if } 0.30 \leq P(\text{risk}|\mathbf{x}) \leq 0.60 \\ \text{High Risk} & \text{if } P(\text{risk}|\mathbf{x}) > 0.60 \end{cases} \tag{1}$$

For all students classified as Medium or High risk, SHAP TreeExplainer [22] generates local feature attributions. Global feature importance is computed as the mean absolute SHAP value across all test samples: $\bar{\phi}_j = \frac{1}{N}\sum_{i=1}^{N} |\phi_j^{(i)}|$. Individual waterfall plots decompose each prediction into per-feature contributions, providing counselors with the specific factors driving a student's risk classification.

IV. Experimental Results

All experiments are conducted on the UCI Dropout dataset (n = 4,424), with the train/val/test split fixed at 70/15/15 using stratified sampling (random seed = 42). The test set ($n = 664$) comprises 451 non-risk and 213 at-risk students, yielding a majority-class baseline accuracy of 67.9%. Results are reported on the held-out test set; variance estimates are obtained via a five-seed stability analysis (Section IV-D).

A. Model Performance at Checkpoint 12

Table I presents the full evaluation metrics for all three models at the Week 12 checkpoint, where the complete feature set is available.

TABLE I: Model Performance at Checkpoint 12 (Test Set)

Model	Acc.	Prec.	Rec.	F1-M	AUC	κ
Logistic Reg.	0.858	0.836	0.845	0.840	0.922	0.680
Random Forest	0.869	0.848	0.854	0.851	**0.925**	0.702
XGBoost	**0.870**	**0.851**	**0.853**	**0.852**	0.920	**0.704**

XGBoost achieves the highest F1-Macro (0.852) and Cohen's κ (0.704), indicating substantial agreement beyond chance [31]. All models substantially outperform the 67.9% majority-class baseline, with even the linear baseline (LR) exceeding 0.84 F1-Macro, demonstrating that the fully leakage-free pipeline—including post-split feature selection—produces consistently strong results. Notably, Random Forest achieves the highest ROC-AUC (0.925), slightly exceeding XGBoost (0.920); this inversion occurs because ROC-AUC evaluates the full ranking of predicted probabilities whereas F1-Macro evaluates performance at a fixed decision threshold, and XGBoost's sharper probability calibration favors the latter.

Fig. 2 displays the ROC curves for the three models at Checkpoint 12, confirming that ensemble methods and the linear baseline achieve comparable discriminative ability on this feature set.

Fig. 3 presents the confusion matrices, illustrating the error distribution across risk categories for each model.

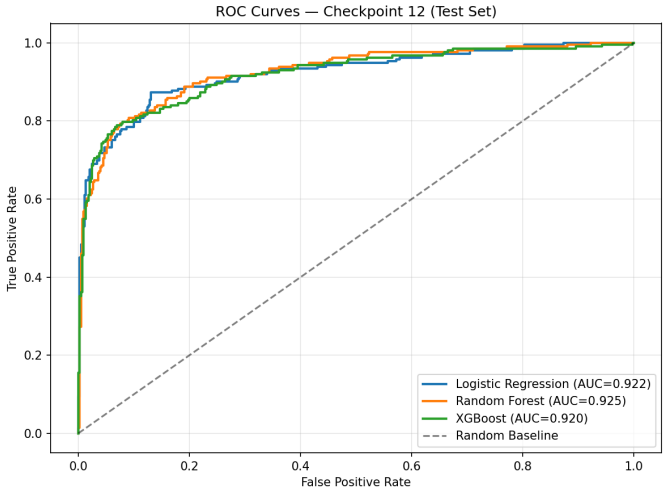


Fig. 2: ROC curves at Checkpoint 12 (test set). All models achieve AUC $\geq$ 0.920, indicating strong discriminative performance.

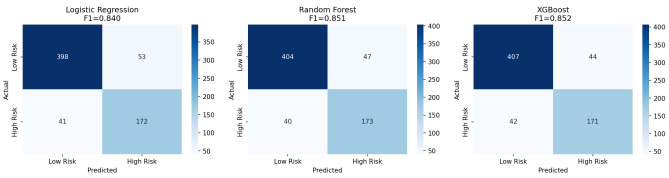


Fig. 3: Confusion matrices at Checkpoint 12 (test set). XGBoost achieves the highest F1-Macro (0.852) with balanced precision and recall across both classes.

B. Temporal Checkpoint Analysis

Table II presents the F1-Macro progression across all three temporal checkpoints, and Fig. 4 visualizes this evolution.

TABLE II: F1-Macro Across Temporal Checkpoints (Test Set)

Model	Week 4	Week 8	Week 12
Logistic Regression	0.735	0.806	0.840
Random Forest	0.715	**0.819**	0.851
XGBoost	0.702	0.813	**0.852**

Three observations emerge from the temporal analysis:

1) **Week 4 baseline:** Using only enrollment and demographic features, all models achieve F1-Macro in the range 0.702–0.735, substantially exceeding the 67.9% majority-class baseline. This represents the lower bound of predictive capability using static pre-admission data alone.
2) **Week 4 $\rightarrow$ Week 8 transition:** The introduction of first-semester academic performance features at Week 8 produces the largest single-step improvement, with F1-Macro increasing by 0.08–0.11 points across all models. This confirms that behavioral academic signals carry substantially more predictive power than static demographic variables.

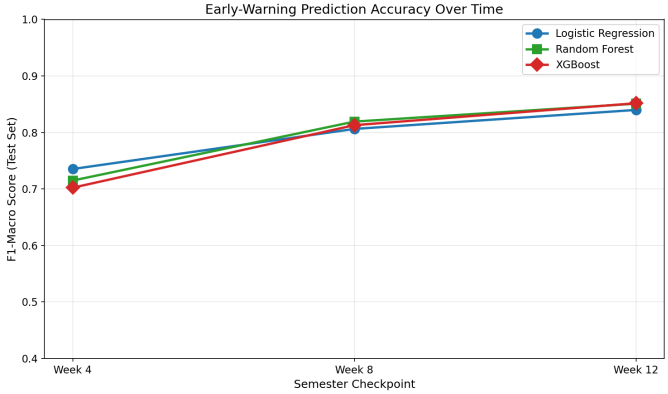


Fig. 4: F1-Macro evolution across temporal checkpoints. The transition from Week 4 to Week 8 produces the largest performance gain ($\Delta \approx +0.08$–0.11), coinciding with the introduction of first-semester academic features.

3) **Week 8 as the operational optimum:** While Week 12 yields the highest absolute performance, Week 8 models achieve F1-Macro $\geq$ 0.806 while providing eight additional weeks of lead time before semester end. This represents the most favorable point on the accuracy–intervention tradeoff curve for practical deployment.

C. SHAP-Based Explainability

1) Global Feature Importance: Fig. 5 presents the global SHAP feature importance for the XGBoost model at Checkpoint 12, computed as the mean absolute SHAP value across all test samples. Fig. 6 extends this analysis with a directional beeswarm plot showing how each feature's value affects risk prediction.

The global analysis reveals that academic performance metrics—particularly second-semester curricular unit outcomes and GPA—are the dominant predictors, followed by first-semester indicators. Enrollment-time features (age, application mode) contribute measurably but with lower magnitude, consistent with the temporal checkpoint results in Section IV.

2) Individual Student Explanations: Fig. 7 presents localized SHAP waterfall plots for a high-risk and a low-risk student from the test set.

These individual explanations transform abstract risk scores into actionable guidance. For example, a counselor reviewing the high-risk profile in Fig. 7(a) can identify that the student's primary risk drivers are poor curricular performance and low GPA, enabling targeted academic support rather than generic advising.

D. Stability Analysis and Computational Cost

To assess the robustness of reported metrics, the full training and evaluation protocol is repeated across five random seeds (42, 123, 256, 512, 1024). Table III presents the mean $\pm$ standard deviation of F1-Macro at each checkpoint.

All models exhibit low variance across seeds ($\sigma \leq 0.008$), confirming the stability of reported results. Logistic Regression, being deterministic given fixed data splits, shows

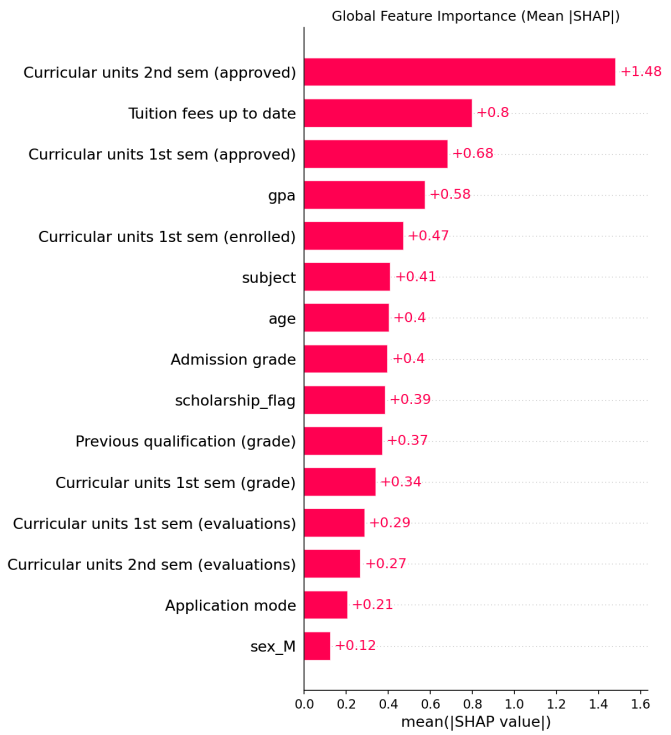


Fig. 5: Global SHAP feature importance (mean $|\phi_j|$). Second-semester curricular performance and GPA dominate, followed by first-semester indicators and enrollment features.

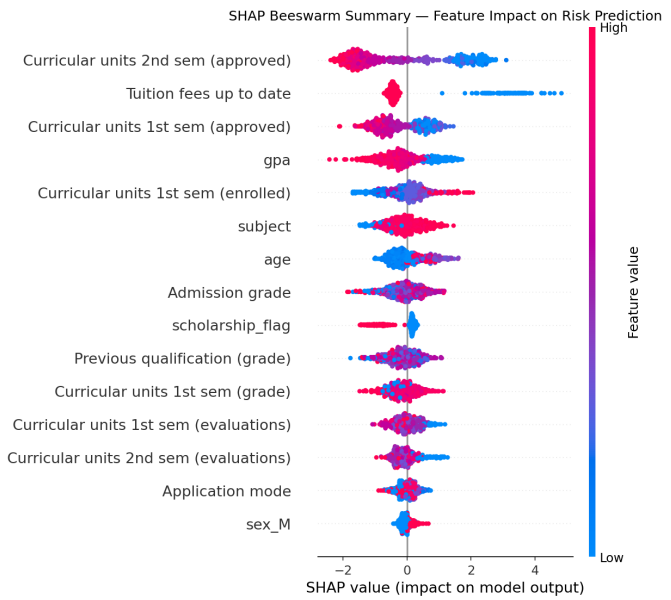
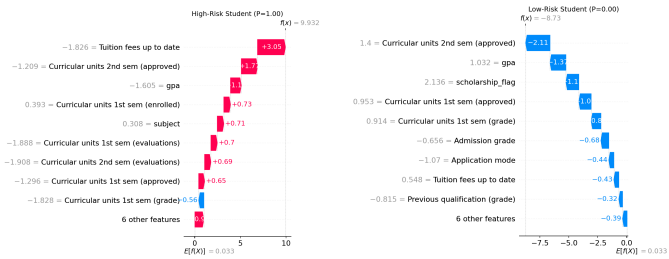


Fig. 6: SHAP beeswarm summary. Red (high feature value) and blue (low value) dots reveal the direction of each feature's impact on risk prediction. Low GPA and high curricular unit failures consistently increase predicted risk.



(a) High-risk student: low approved curricular units and poor GPA drive the elevated risk score.

(b) Low-risk student: strong GPA and high curricular unit completion serve as protective factors.

Fig. 7: Localized SHAP waterfall plots. Each bar quantifies a single feature's contribution to the individual prediction, enabling counselors to identify specific intervention targets per student.

TABLE III: Multi-Seed Stability Analysis: F1-Macro (Mean $\pm$ Std, n=5 Seeds)

Model	Week 4	Week 8	Week 12
Logistic Reg.	0.735 ± 0.000	0.806 ± 0.000	0.840 ± 0.000
Random Forest	0.721 ± 0.008	0.821 ± 0.002	0.852 ± 0.002
XGBoost	0.698 ± 0.001	0.820 ± 0.004	0.847 ± 0.005

zero variance. The 5-fold cross-validation F1-Macro on training data further corroborates test-set performance: XGBoost achieves 0.886 ± 0.008, Random Forest 0.882 ± 0.008, and Logistic Regression 0.860 ± 0.008 at Week 12.

Table IV summarizes computational costs. All experiments are conducted on a consumer-grade workstation (Intel Core i7, 16 GB RAM, no GPU acceleration).

TABLE IV: Training Time per Model at Checkpoint 12

Model	Train (s)	Inference (s)
Logistic Regression	0.02	< 0.01
Random Forest	0.87	< 0.01
XGBoost	0.54	< 0.01

The sub-second training times confirm that ProGrade is deployable on institutional hardware without specialized infrastructure. End-to-end pipeline execution (feature engineering through evaluation across all three checkpoints) completes in under 30 seconds.

V. Discussion

A. Addressing the Research Questions

RQ1 (Architecture and Reproducibility): The Medallion pipeline provides three key advantages over flat-file approaches. First, immutable Bronze storage preserves raw data provenance for audit. Second, the Silver star-schema enforces referential integrity and enables consistent cross-dataset joins. Third, the Gold data mart decouples analytical outputs from the ML pipeline, allowing BI dashboards and counselor reports to operate independently of model retraining cycles. These

architectural decisions directly support reproducibility, as any downstream analysis can be traced to its source records.

RQ2 (Optimal Checkpoint): The temporal analysis reveals Week 8 as the most operationally valuable checkpoint. The Week 4 $\rightarrow$ Week 8 transition yields the largest marginal gain (ΔF1 $\approx$ $+0.08$–0.11), driven by the availability of first-semester academic features. The subsequent Week 8 $\rightarrow$ Week 12 gain is more modest (ΔF1 $\approx +0.03$), while the lead time decreases by four weeks. For institutions prioritizing early intervention, Week 8 provides an F1-Macro exceeding 0.80 with eight weeks remaining for counselor-led remediation. The five-seed stability analysis (Table III) confirms that this conclusion is robust across random seeds ($\sigma \leq 0.004$ at Week 8).

RQ3 (Explainability): The combination of global SHAP importance rankings (Fig. 5) and individual waterfall plots (Fig. 7) addresses the trust deficit identified in prior work [15]. Rather than presenting a binary risk label, ProGrade provides counselors with a ranked list of contributing factors per student, mapped to the three-tier classification in Eq. 1. This supports differentiated interventions—e.g., academic tutoring for students with low GPA attribution versus attendance monitoring for absence-driven risk.

B. Comparison with Prior Work

Table V positions ProGrade relative to recent studies on the same or comparable datasets.

TABLE V: Comparison with Related Work on Student Risk Prediction

Study	Metric	Score	Temporal	XAI
Realinho et al. [1]	Accuracy	0.79	No	No
Martins et al. [21]	F1-Macro	0.83	Yes	No
Ahmad et al. [32]	Accuracy	0.88	No	No
Alwadi et al. [16]	Accuracy	0.85	No	Yes
ProGrade (ours)	**F1-Macro**	$\mathbf{0.852 \pm 0.005}$	**Yes**	**Yes**

Note: Direct numerical comparison across studies is approximate, as reported metrics differ (Accuracy vs. F1-Macro). ProGrade reports F1-Macro with five-seed variance estimates to ensure methodological transparency.

ProGrade is, to our knowledge, one of the first studies to unify a Medallion warehouse architecture, systematic temporal checkpoint evaluation, fully leakage-free preprocessing (including post-split feature selection), and SHAP-based per-student explanations within a single framework.

C. Limitations and Future Work

Several limitations warrant discussion:

1) **Simulated temporal masking:** The checkpoint protocol simulates temporal availability by masking features in a static dataset. Authentic deployment requires integration with institutional Learning Management Systems (LMS) for real-time data feeds.
2) **Single-institution datasets:** While three datasets are unified in the warehouse, they originate from similar educational contexts (Portuguese institutions). Cross-institutional transfer learning [24] remains unexplored.
3) **Binary risk classification:** The current pipeline predicts a binary outcome (at-risk / not-at-risk). Multi-class prediction (Dropout / Enrolled / Graduate) [21] may provide more granular intervention guidance.
4) **Feature modality:** The current system operates exclusively on structured tabular data. Future work could integrate NLP-based analysis of student communication logs and LMS engagement traces as supplementary behavioral signals [23].
5) **Fairness analysis:** We do not formally assess model fairness across demographic subgroups. Given the inclusion of sensitive attributes (gender, socioeconomic status), future work should employ established fairness metrics—including equalized odds, demographic parity, and calibration across groups—as a prerequisite for institutional deployment [24].

VI. Conclusion

This paper presented ProGrade, an end-to-end early warning framework that addresses four persistent gaps in educational data mining: the absence of reproducible data architectures, data leakage in preprocessing, temporal mismatch between prediction and intervention, and insufficient model transparency. By integrating a Medallion data warehouse (Bronze $\rightarrow$ Silver Star-Schema $\rightarrow$ Gold Data Mart) with fully leakage-free preprocessing (including post-split feature selection), temporal checkpoint modeling, and SHAP-based explainability, ProGrade provides a principled foundation for institutional deployment.

Our experimental evaluation on the UCI Dropout dataset ($n = 4{,}424$) demonstrates that XGBoost achieves an F1-Macro of 0.852 ± 0.005 and ROC-AUC of 0.920 at Week 12 (five-seed stability analysis), with Week 8 models attaining F1-Macro ≥ 0.80 while preserving eight weeks of intervention lead time. All models exhibit low variance across seeds ($\sigma \leq 0.008$), confirming result robustness, and sub-second training times confirm deployability on consumer-grade hardware. SHAP TreeExplainer generates transparent, per-student explanations that enable academic counselors to identify specific risk drivers and design targeted intervention strategies.

Future work will focus on real-time LMS integration, multi-class outcome prediction, cross-institutional transfer learning, and formal fairness auditing using equalized odds and demographic parity metrics, to advance ProGrade toward production deployment in diverse educational settings.

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